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Forming a Database of Damaged and Destroyed Research Infrastructure in Ukraine

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Summary

As of November 2022, there was no consistent effort to collect information on damaged and destroyed research infrastructure *(RI)*. Different institutions used different methods (or none) to collect the data. Often these databases were incomplete. This made it difficult to assess damages done to the RI and estimate the reconstruction costs.

The topic of crisis management and disaster management has been extensively studied in the global literature. While conceptually similar studies, as well as occasional scientific reports and news articles have been identified in the literature, the focus is on the general infrastructure damaged by man-made and natural disasters rather than armed conflicts and wars. Studies on the damages done to the RI are virtually nonexistent.

After reviewing the literature, official sources, media reports, and information obtained directly from Ukrainian scientists, a sample of eight scientific organizations and higher education institutions *(HEIs)* were selected. The RI of all institutions in the sample suffered damage and destruction. Equipment, archives, collections, libraries, physical and electronic data repositories, as well as the supporting infrastructure necessary for the functioning of research equipment, were damaged, destroyed, or looted. Two of the studied organizations are currently in the process of assessing the damage. Persistent problems with water and electricity supply continue to cause damage to RI or prevent its reconstruction. The long-term impact of these factors on RI is unknown.

This white paper seeks to address this issue in two ways. Firstly, it presents research on the challenges of collecting information about damages and destructions of RI during times of war. Secondly, based on an in-depth study of individual cases, it systematizes data on damaged and destroyed RI in the form of a table. The table is a proposal for the further development of a database for the systematic collection of information on damaged and destroyed RI. The white paper also includes recommendations on the creation and maintenance of such a database.

The working group conducted a survey to develop a table of damaged and destroyed RI. To supplement and clarify the information obtained via the survey, semi-structured in-depth interviews were conducted with managers or authorized representatives of selected institutions. The questionnaire and interview consisted of three parts: 1 — General information, 2 — Information about the damage done to the RI, and 3 — Fundraising and the future. Given the short time frame of the project

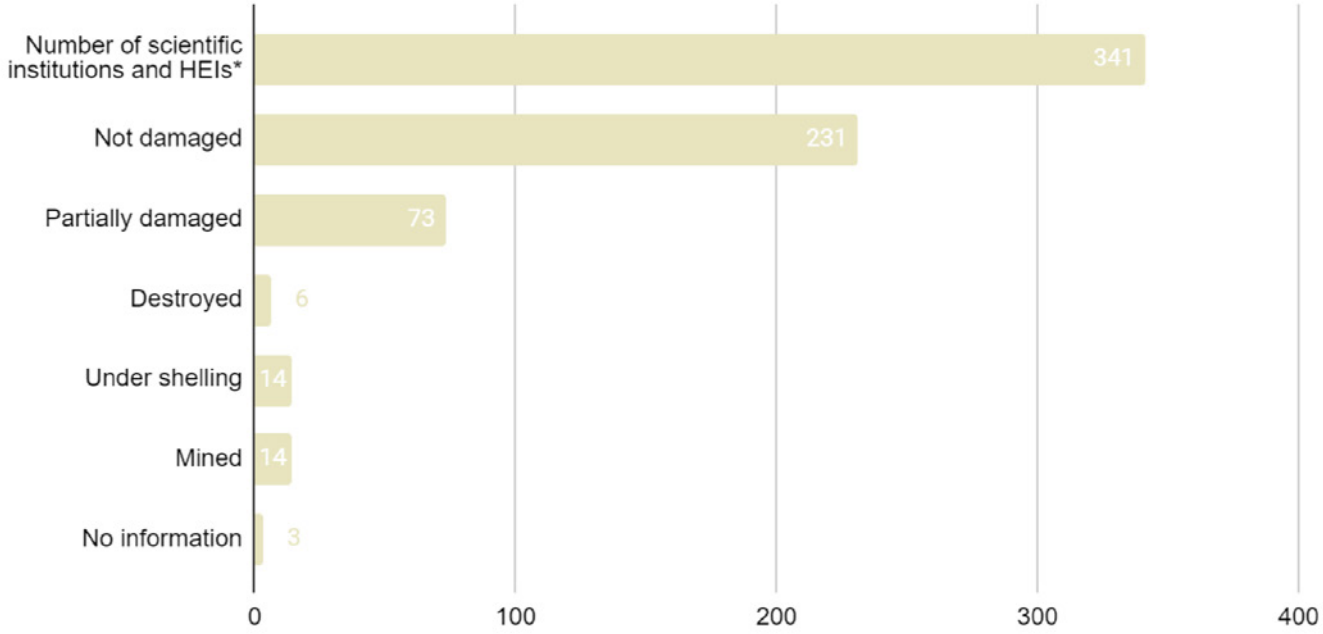
and the limited human resources, the study was focused on material or physical damages to the RI. However, it is equally timely and relevant to investigate the non-material damages (human losses, lost experience, unfinished work, etc.). We recommend collecting data for the study of these and other questions now.

In summary, the working group provided recommendations on creating a database of damaged and destroyed RI, its administration, and the dissemination of information. Such a database must become a trusted and influential resource where comprehensive information about affected RI can be found. Similar initiatives and projects that monitor losses inflicted on Ukrainian science should be consolidated to avoid duplication of efforts. The database on RI losses must be regularly updated in order to provide the most complete and up-to-date information. Funds must be allocated for the creation and administration of this database to ensure its efficient and sustainable functioning. The database should not be managed on a voluntary basis. It is also essential to inform the national and international media, social networks, and scientific platforms about the database. The functioning of science and the losses inflicted on RI during the war should be discussed at international events and scientific fora to engage the Ukrainian and global community in a deeper discussion about the impact of war on science and prospects for future development.

1. Research problem

As of January 2023, because of the armed aggression of the Russian Federation against Ukraine, about 110 scientific institutions and higher educational institutions *(HEIs)* are known to have been damaged (see diagram 1). In addition, about 28 scientific institutions and HEIs could not determine the extent of the damage since they were shelled, mined, or located in the occupied territory[1]. A report published in October 2022 by the Ministry of Education and Science of Ukraine *(MESU)*[2] and international partners noted that the number of surveyed HEIs that reported infrastructure damage due to the war was around 165. Among the surveyed institutions, 81 HEIs, or 11%, reported damage to laboratory equipment, and 12% reported damage to the information and communication infrastructure.

Diagram 1. The state of Ukrainian scientific institution and HEIs during the war



Source: Reference information provided by MESU (December 2022).

** This number includes scientific institutions and HEIs that responded to the survey by MESU. The total number of such institutions in Ukraine exceeds 341.*

1 Reference information provided by the Ministry of Education and Science (December, 2022)

2 MoES, World Bank and Ukraine Education Cluster (October, 2022): ukraine_higher_education_assessment_report_24_october_2022 (1)

The problems and challenges faced by Ukrainian science before the start of the full-scale war on February 24, 2022, were complicated by the physical destruction and significant damage done to buildings and other property of scientific institutions and HEIs, as well as collateral damage caused by power outages, damage to systems water supply and heating. The loss, displacement, and incapacitation of researchers as well as the abductions of scientists in the occupied territories, have become serious challenges for Ukrainian science.

As of November 2022, data collection on the damaged and destroyed research infrastructure *(RI)*[3] has been carried out unsystematically by various institutions and using different methodologies (often none at all). Such databases were often incomplete. Despite the existence of an order approved by the Cabinet of Ministers of Ukraine[4] on determining the damage caused to Ukraine as a result of the armed aggression of the Russian Federation, this document does not include RI as a separate item. This makes it impossible to apply a standardized approach to determining the damage and losses, as well as their evaluation for reconstruction purposes. This is not surprising, as the issue of evaluating both material and non-material losses from the perspective of lost scientific opportunities is not a well-researched phenomenon. Given the short time frame of the project and limited human resources, the study was focused on material or physical losses inflicted on RI. However, it is no less urgent to investigate intangible losses (casualties, lost experience, incomplete work, etc.). This may be the subject of the next study.

3 RI is a set of tools, resources, and related services used by the scientific community to conduct research at the highest level, covering the most important objects of scientific equipment and instrumentation or sets of instruments, knowledge-based resources (collections, archives, depositories or scientific information data banks), infrastructure based on communication technology (grid, computers, software and network communication), and other unique structures. RI can be locally situated, virtual, or distributed (organized network of resources), state-run or private. RI can be part of international networks. (Ministry of Education and Science).

4 Verkhovna Rada of Ukraine On approval of the Procedure for Determining the Damage and Losses Caused to Ukraine as a Result of the Armed Aggression of the Russian Federation. Resolution № 326 of 20.03.2022.

2. Research task

Within the framework of the *Science at Risk project*, a working group was formed with the task of (1) researching the issue of collecting information on damage and destruction of RI during the war and (2) based on the in-depth study of individual cases, systematizing data on damaged and destroyed RI in the form of a table. This table is attached to this report as a proposal for the further development of a database for the systematic collection of this information. Recommendations for the formation, administration, and use of the database, as well as communication, are included in the last section.

3. Literature review

The initial step in the research process involved searching for literature and examples documenting the damage incurred by scientific institutions and HEIs due to war-related destruction. The topics of crisis and disaster management have been widely studied in the literature. More general and conceptual studies in the field of disaster management[5] have been identified, along with individual reports by scientific communities[6] and news coverage in media outlets[7], primarily in the context of disaster incidents resulting from man-made and natural catastrophes. As an example, Drennan et al.[8] identify the following categories of consequences that crises may have on public organizations and HEIs:

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5 Aronsson-Storrier, M., & Dahlberg, R. (Eds.). (2022). Defining Disaster: Disciplines and Domains. Edward Elgar Publishing;
Dahlberg, R., Rubin, O., & Vendelø, M. T. (Eds.). (2015). Disaster research: Multidisciplinary and international perspectives. Routledge.

6 National Academies of Sciences, Engineering, and Medicine. (2017). *Strengthening the disaster resilience of the academic biomedical research community: Protecting the nation's investment.*
Riedlmayer, A. J. (2007). Crimes of War, Crimes of Peace: Destruction of Libraries during and after the Balkan Wars of the 1990s. Library Trends, 56(1), 107–132.
Zgonjanin, S. (2005). The prosecution of war crimes for the destruction of libraries and archives during times of armed conflict. Libraries & Culture, 128–144.

7 Stafford, N. (21 March, 2012). Quake-hit Japanese universities move on. Retrieved from https://www.chemistryworld.com/news/quake-hit-japanese-universities-move-on/3003355.article
Gottlief, J., and Beatriz Cholo, A. (25 July, 2001). Explosion in UC Irvine Science Lab Caused Less Damage Than Feared. Retrieved from https://www.latimes.com/archives/la-xpm-2001-jul-25-me-26368-story.html

8 Drennan, L., McConnell, A. & Stark, A. (2015). Risk and Crisis Management in the Public Sector. 2nd Edition. New York, USA.

- Human losses: the death and injury of workers, as well as psychological trauma.
- Infrastructural losses: the destruction of buildings, intellectual property, library resources, laboratories, office equipment, archives, etc.
- Economic losses: including job losses, reduced income, and other negative economic impacts.
- Loss of legitimacy: resulting in decreased trust and confidence in the organization.

During the study of practices and precedents, several national initiatives and projects were identified that map the damaged civil infrastructure[9], schools[10], and cultural heritage[11]. The Kyiv School of Economics is developing a methodology for assessing the inflicted economic damages[12].

Unfortunately, the issue of damages to the RI was not specifically addressed in these studies and projects. This again shows the importance and value of further research on this topic.

4. Research methods

After reviewing the literature, government sources, media reports, and information obtained directly from scientists, the working group formed a sample of eight scientific institutions and HEIs whose RI incurred damages due to the war. The number of members of the working group (four) and the time frame of the project (December 2022 — January 2023) allowed for optimal and high-quality processing of eight case studies using a survey and in-depth semi-structured interviews.

The process of obtaining consolidated information on the damage and destruction of research institutions and HEIs proved to be complicated, as the document recording the state of activity of these institutions was received from the relevant government organizations only in January 2023 (with updates in February 2023), after the working group has developed the white paper. Additionally, the information was not comprehensive, as it was based only on data provided by research institutions and HEIs that responded to the survey conducted by MESU in the fall

9 Map of the Reconstruction and Damages [Карту Відновлення та Руйнувань].

10 Education under threat [Освіта під загрозою].

11 Destroyed cultural heritage of Ukraine.

12 The total damage caused to Ukraine's infrastructure is more than $ 127 billion — KSE Institute's report as of September 2022 — Kyiv School of Economics

of 2022. Moreover, since Ukraine is still in a state of war, the information obtained before January 2023 is not exhaustive, which is one of the limitations of this study. During the writing of this white paper, there were massive shelling incidents, which led to the destruction of new research institutions and HEIs. Also, valuable information for this study, such as reports on further destruction[13], continues to emerge in the public sources, which were previously unavailable, and the working group was thereby unable to process it.

4.1 Research design

To create a table of damaged and destroyed RI, the working group conducted a survey via *Google Forms* of eight institutions in December 2022. To supplement and refine this information, in January 2023, the group members conducted semi-structured in-depth interviews with the directors or authorized representatives of the selected institutions.

The questionnaire consisted of three parts: 1—General information, 2—Information about the damages done to the RI, and 3—Fundraising and the future. The interviews had the same structure. If the answers in the questionnaire were unclear or insufficient, the interviewers asked for clarification.

Most interviews lasted up to an hour. The interviewers understood the unpredictability of the situations (air raids, power outages) in which the respondents found themselves; thus, the interviewers tried to adapt to these circumstances. Various platforms, such as *Zoom, WhatsApp, Viber,* etc., were used for conducting interviews. With the respondents' permission, the interviews were recorded for further qualitative information analysis.

The data obtained from the survey responses were transferred to a table, which is included in the appendix. The information obtained during the in-depth interviews served as the basis for the analysis (section 5) and the development of recommendations (section 7). Quotes from some of the interviews are provided to illustrate the scope of the problem of damage and destruction of RI.

13 Kyiv Taras Shevchenko National University.
Report about the damages caused by the Russian aggression against Ukraine.

4.2 Sample

The following cases were included in the sample:

1. State Scientific Institution «Institute for Single Crystals» of National Academy of Sciences of Ukraine, Kharkiv
2. Institute of Radio Astronomy of the National Academy of Sciences of Ukraine, Kharkiv
3. Institute of Agricultural Microbiology and Agro-Industrial Production of the National Academy of Agrarian Sciences of Ukraine, Chernihiv
4. M. M. Hryshko National Botanical Garden of the National Academy of Sciences of Ukraine, Kyiv
5. Institute of Bio-Stem Cell Rehabilitation of the Ukrainian Association of Biobanks, Kharkiv
6. Sumy National Agrarian University, Sumy
7. V. N. Karazin Kharkiv National University, Kharkiv
8. State-specialized enterprise Ecocenter, Chornobyl.

The sample was compiled based on several factors, namely the location regarding the geography of active hostilities during 2022, subordination, ownership form and status, disciplinary profile, as well as the availability of information. Unfortunately, many institutions in the east and south of Ukraine are still in occupied territories or are under shelling. They were not included in the sample due to the inability to obtain data. The sample consists of scientific institutions and HEIs from the territories of the north and east of Ukraine that were not occupied or were de-occupied and where information is publicly available.

5. Presentation of analytical work results

All institutions in the sample suffered some degree of loss of RI. Equipment, archives, collections, libraries, data repositories (physical and electronic), as well as supporting infrastructure necessary for research equipment operation were damaged, destroyed, or looted. In addition, the servers of some organizations were compromised, and other communication technologies were damaged. In all but one institution, RI requires reconstruction.

A colleague called me and said that the Russians had entered through a side entrance, there were two military vehicles, and they were taking things out, starting to loot the laboratory. We were mentally prepared that the laboratory was gone. After the de-occupation, on April 4, we returned there. When I brought in the staff, I gave them the task to go through the laboratory and take photos and videos of everything. From the servers and hundreds of computers, nothing left. It was a complete horror; everything was destroyed and looted. So as of April 4, 2022, the occupiers completely put the laboratory out of operation.
(Interview, January 2023, Head of the State specialized enterprise Ecocenter).

Four institutions have fully resumed their activities, two have partially resumed, one is currently restoring scientific activities, and one has suspended its work. In the case of the biobank in Kharkiv, some collections and equipment were relocated to other European countries. At the same time, the RI continues to operate according to the plan, and additional activity directions are being developed. The Institute of Radio Astronomy in Kharkiv had to move some of its equipment to other buildings in the institute to continue some of its activities while full demining and repair work is being carried out at the observatory site. In addition, the institute is continuing the planned modernization of some equipment and working on an international project to create a new instrument for joint use—the Giant Ukrainian Radio Telescope—based on the existing telescope.

The total damage caused by the occupied Russian forces to eight Ukrainian scientific institutions is approximately 5 billion UAH (almost 140 million USD). The losses of V. N. Karazin Kharkiv National University alone are the largest, reaching 4.5 billion UAH (more than 120 million USD).

— You mentioned an estimated loss of RI of 4.5 billion UAH. Is that a typo? 4.5 billion?

— No, unfortunately, that is not a typo. Only in Kharkiv, 19 facilities have been destroyed/damaged, not including facilities in the region (practice bases, observatories, laboratories, etc.). These are actually quite optimistic calculations regarding the total losses.
(Interview, January 2023, Executive Director of the charitable organization of V. N. Karazin Kharkiv National University).

The conditions and methodology for assessing the damages and necessary reconstruction of the RI of the eight institutions differed. Some stated that the practice of organizing an assessment of material/physical damages to infrastructure was absent, so it was necessary to create it on

the spot. No standard protocol was established. The uniqueness of scientific collections and equipment complicates assessing their value. In the case of V. N. Karazin Kharkiv National University, the assessment is further complicated by the damaged and destroyed unique historical archives.

> *We have never been in these conditions before. I tried to search for experience in the world, maybe someone has already dealt with this, but I did not find any answers.*
> *(Interview, January 2023, Executive Director of the Charitable Organization of V. N. Karazin Kharkiv National University).*

> *No one was ready for this. On February 24, we were preparing for a seminar. [...] As for the recording of damages [during the period of hostilities in the city of Chernihiv], it was irrelevant and not discussed because the question was how to survive, not how to record. Some units carried out «initial» recording—short notes in the duty log. [...] Full recording took place several weeks after the retreat of the Russians from under Chernihiv and the lifting of the siege of the city and cessation of shelling when law enforcement agencies arrived at our request and took testimony from employees who witnessed property damage. Later, another group arrived and made a video recording. [...] even the restoration of water, gas, and electricity supply was not as relevant as the urgent issue of demining research sites because field research had to be carried out in the spring. Everything happened quite spontaneously, but the process of RI damage assessment appeared before me only in practical terms from your survey because, at that time, we were solving the most urgent needs.*
> *(Interview, January 2023, Director of the Institute of Agricultural Microbiology and Agro-Industrial Production of the National Academy of Agrarian Sciences of Ukraine).*

Some institutions have not yet conducted an assessment because some of the areas are still mined. Representatives of two organizations noted that their experience suggests that the territories of scientific institutions are not a priority for demining by units of the State Emergency Service of Ukraine. Several institutions have also reported that their equipment continues to suffer damage because of frequent power outages, thus prolonging the assessment of damage and reconstruction and potentially changing the cost.

Here are the key results of the actions taken regarding the assessment of damage and reconstruction of RI (in no particular order):

- Creation of an expert commission capable of inspecting the infrastructure and determining the cost of damage and reconstruction;
- Preparation of reports on losses;

- Conducting an inventory of infrastructure;
- Performing an audit with the involvement of external experts;
- Establishing commercial or market prices for the equipment.

One of the organizations described the process of assessing damages as follows:

> *The work of the experts involved studying the technical documentation that we had, as well as inspecting all the buildings. The head of the economic department went with them, and they conducted photo documentation of our five buildings [...]. Then, based on the study of technical documentation and the actual condition of the buildings, they began the assessment. It is very important to us, and we reminded them of it constantly—in November, in December. In our latest communication, they informed us that they should finish around mid-January. As for the method they use—fair market value, residual book value, or something else—unfortunately, I do not have this information. I assume that we are not the first ones to turn to them, and of course, in the city of Chernihiv and the region, many objects have suffered greater damages [...]*
> *(Interview, January 2023, Director of the Institute of Agricultural Microbiology and Agro-Industrial Production of the National Academy of Agrarian Sciences of Ukraine).*

More than half (5) of the institutions had formulated a fundraising strategy and a reconstruction plan, outlining steps to raise funds to restore the research institution.

> *We have developed a reconstruction plan, and we have already completed some of the items on that plan. We met with representatives of the European Commission and tasked them with restoring the laboratory to its pre-war condition. They are helping us with this. We inform them of what we have already checked and what we have not been able to check, as well as the reasons why, and what remains to be done to restore the research institution*
> *(Interview, January 2023, Head of the State specialized enterprise Ecocenter).*

Among the actions taken to restore infrastructure were the following:

- Identification of potential donors and reaching out to them through various channels (social media, personal contacts, etc.);
- Appeal for support to institutions to which the organization is subordinate (government agencies, ministries);
- Dissemination of information about the extent of the damage through the media;

- Opening bank accounts (in different currencies) for fundraising;
- Submission of applications for international grants.

The system of subordination of some institutions allowed for a more structured approach to seeking help with fundraising. In-depth interviews revealed that the special status of certain scientific institutions and their unique value to the country (and, in some cases, to all of Europe) enabled the institution to leverage the necessary levers of influence for fundraising for reconstruction.

> *[We are] the only laboratory in Ukraine that can determine all these parameters and comply with IAEA recommendations for the characterization of radioactive waste. This is an important laboratory in terms of safety not only for the country but also for Europe. Therefore, our strategy for fundraising is to approach a company, and they either respond to us or not. Then we involve the government agency. If that doesn't help, we involve the European Commission agency.*
> *(Interview, January 2023, Head of the State specialized enterprise Ecocenter).*

Most institutions (6) received assistance for the maintenance and reconstruction of their RI. Mainly international partners and some domestic donors provided assistance. However, it should be noted that the aid did not cover all the losses of the institutions. The assessment of the damages did not include the human resources and the time they spent on research, as well as their expertise..

> ***Question from the survey:** Please indicate where information about damage to RI can be found on your organization's website.*
> ***Answer:** Unfortunately, such information is not available on our website, as our IT department was hit the hardest. Our specialists died.*
> *(Interview, January 2023, Director of the Institute of Bio-Stem Cell Rehabilitation of the Ukrainian Association of Biobanks).*

Some respondents mentioned during in-depth interviews that existing regulations on the use of budget funds do not allow for the use of any special funds for the preservation/reconstruction of RI, except for charitable donations. Moreover, treasury structures usually simply block transactions, taking on functions that are not typically their responsibility. It is worth noting that more than half of the institutions (5) shared information about the damaged and destroyed RI in English. This information was posted on the websites of the institutions themselves, as well as on the websites of partners and in the media. All institutions, except one, have

an authorized contact person who communicates with potential donors in English.

Although this work was focused on researching the material losses incurred by RI, several respondents mentioned during interviews the challenges that scientific institutions and higher education institutions might face in restoring scientific activity. Firstly, it is impossible to restore the lives of the lost employees, whose experience was valuable and sometimes irreplaceable for certain research. Secondly, there is also a risk that some employees may not return due to emigration, resulting in a shortage of qualified personnel. After the end of the war, it will become a priority to attract new personnel to these scientific fields in Ukraine, as well as to provide training on the use of RI. Finally, respondents are unsure if it will be possible or worthwhile to restore RI to its pre-war state. Some scientific equipment is outdated and needs to be updated. Therefore, it is worth assessing losses in the context of further scientific development. Strategies for restoring RI in scientific institutions and HEIs should reflect these perspectives.

6. Conclusions

In Ukraine, there is still no unified methodology for collecting data on damaged and destroyed RI. Neither the members of the working group nor the respondents have identified any precedents for documenting RI damage caused by military hostility, armed conflict, or war elsewhere in the world. The working group searched for information published in Ukrainian and English. It is not excluded that documentation of some precedents could be conducted in other languages, so in the future, it is advisable to expand the search to other languages (of countries where military hostilities have been taking place or had taken place).

The methodology proposed by the working group for in-depth work with individual cases to form a structure of a database of damaged and destroyed RI focused on assessing the losses inflicted on various types of RI during the war. However, this approach could also be useful for assessing losses to other types of infrastructure, such as cultural heritage sites or other types of infrastructure affected by disasters. Moreover, the discussion regarding the evaluation of losses of the RI with cultural and historical value (such as a unique collection of beneficial bacteria of the National Academy of Agrarian Sciences, which was formed over 60 years and has the status of national heritage; or valuable historical books in the

university libraries) should be conducted in collaboration with experts in cultural and historical heritage preservation.

One weakness of the proposed method is that the information obtained by the working group from respondents was mostly focused on estimating direct material damages. Still, such losses are often closely linked to intangible losses (loss of life, time, damaged reputation, lost opportunities or potential, etc.). Therefore, developing a methodology for assessing intangible losses is also needed. Involving experts in economics, finance, and legal studies will help fill the gaps identified in this work.

It is also important to consider indirect losses, such as the loss of income from planned contractual work that could not be carried out at all or on time due to the lack of electricity/water/damaged equipment or the inability to maintain other statutory activities of institutions and organizations (providing services, promoting science, educational programs, etc.).

Therefore, the results of the working group's activities at this stage are not a guide for general use but rather a starting point that has fulfilled its task of (1) investigating the problem of damage and destruction of RI based on a small sample of cases, and thus (2) starting to develop a methodology on forming the database. The next section provides recommendations on the formation, administration, and use of the database, as well as informs about it.

7. Recommendations

As a summary of the information obtained from the experience of eight case institutions, the working group has prepared recommendations regarding database formation, administration, use of the database, and communication about the database. This database should become a practical and functional tool that will be known both to the Ukrainian and global communities. At this stage, the recommendations are provided regarding the approach to working with the database without considering the institutional capabilities of individual organizations for its implementation as a separate project.

7.1 Formation of the database

1. Coordinate the efforts of projects and initiatives that carry out similar work (see section 3) on assessing the damages of RI.

2. Agree on the methodologies for populating the database by coordinating among government agencies.
3. Determine who has the right to provide this information. The database should cover various types of research institutions for better coordination of efforts to restore RI. The database should include an address to the heads of scientific institutions and HEIs, explaining, among other things, the procedures for submitting data about destroyed and damaged RI. These procedures should also specify that only one person authorized by the institution can enter such data.
4. Engage in accounting for damaged and destroyed RI in the national electronic scientific information system «URIS» of the State Scientific and Technical Library of Ukraine.
5. Form the entire database in two language versions (Ukrainian and English) for the possibility of dissemination of this database with partners and interested parties.
6. The «discipline» or «type of infrastructure» columns should include the most detailed classification possible. This will provide an opportunity to direct the search and build partnerships between Ukrainian and international communities.
7. Add a feature of visualization of damages for effective communication of the scale and urgency of the problem of destruction of RI.
8. Include separate columns for «amount of damages» and «amount needed for reconstruction» with explanations of which RI is damaged and why the exact amount is needed.

7.2 Administration of the database

1. Implement a sustainable approach to the development and use of the database as a resource: the database should be as user-friendly as possible for its visitors (potential donors, researchers, journalists). Visitors to the database should be allowed to apply filters to search for information («location,» «disciplinary profile,» «type of RI»).
2. The database should be maintained regularly. Separate funds and personnel should be allocated for its administration (not on a voluntary basis).
3. The date of the last update of the information in the database should be indicated, as the data may constantly change.
4. The survey for filling the database should remain open or partially restricted, with the possibility of adding information.
5. The database should be updated until the end of the war, as well as in

the first months after its termination. This will allow for the most comprehensive assessment of damages to the RI of scientific institutions and HEIs.

7.3 Informing about the database

1. The database must be well-known, authoritative, and effective. It is important for every institution whose RI has been affected to know about its existence and have the opportunity to independently enter information about the damage into this database.
2. In Ukraine, it is important to spread information about the database through the media, social networks, partner websites, and state organizations to which the institutions are subordinate.
3. For the international community — disseminate information through the media, Science Business journal, social networks, and on project websites that have experience in mapping RI in Europe (and beyond) and forming catalogs of RI services for users from the scientific community and industries.
4. Convey the relevance of the problem to the European Strategic Forum for Research Infrastructures *(ESFRI)* and other international forums and platforms for discussions on scientific policy and RI.

Appendix: Database of the damaged Ukrainian research infrastructure

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